DMA LAB-7 R BASICS

LOADING THE DATA

```
> df <- read.csv("D:/fruits.csv")
> head(df)
  weight size color_score Fruit_label

1    150  7.5        0.50       lemon
2    170  7.5        0.50       lemon
3    190  8.0        0.74      orange
4    210  8.5        0.75      orange
5    230  8.5        0.75      orange
6    250  8.5        0.80       Apple
```

DESCRIPTION OF DATA

```
> str(df)
'data.frame':  23 obs. of  4 variables:
 $ weight     : int  150 170 190 210 230 250 270 290 165 185 ...
 $ size       : num  7.5 7.5 8 8.5 8.5 8.5 8.5 9 7.5 7.5 ...
 $ color_score: num  0.5 0.5 0.74 0.75 0.75 0.8 0.8 0.8 0.5 0.5 ...
 $ Fruit_label: chr  "lemon" "lemon" "orange" "orange" ...
```

CLASS DATATYPE

```
> class(df$color_score)
[1] "numeric"
```

LENGTH OF A CLASS

```
> length(df$color_score)
[1] 23
```

DIMENSIONS OF DATA

```
> dim(df)
[1] 23  4
> ncol(df)
[1] 4
```

CLASS NAMES / COLUMN NAMES

```
> names(df)
[1] "weight"      "size"        "color_score" "Fruit_label"
```

CENTRAL TENDENCY

```
> mean(df$weight)
[1] 231.3043
> median(df$weight)
[1] 230
```

MEASURES OF DISPERSSION

```
> sd(df$weight)
[1] 45.95495
> var(df$weight)
[1] 2111.858
> cov(df$weight, df$size)
[1] 18.20158
> range(df$weight)
[1] 150 305
```

SUMMARY OF A CLASS

```
> summary(df$weight)
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max. 
  150.0   195.0   230.0   231.3   267.5   305.0 
> summary(df)
     weight           size        color_score     Fruit_label       
 Min.   :150.0   Min.   :7.500   Min.   :0.5000   Length:23         
 1st Qu.:195.0   1st Qu.:7.500   1st Qu.:0.5000   Class :character  
 Median :230.0   Median :8.000   Median :0.7600   Mode  :character  
 Mean   :231.3   Mean   :8.152   Mean   :0.7191                     
 3rd Qu.:267.5   3rd Qu.:8.500   3rd Qu.:0.8100                     
 Max.   :305.0   Max.   :9.000   Max.   :0.9200                     
```

TABLE

```
> table(df$weight>200)

FALSE  TRUE 
    7    16 
> table(df$Fruit_label==("Apple"))

FALSE  TRUE 
   16     7 
```

IS NA TO FIND NULL VALUES

```
> is.na(df$weight)
 [1] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE
[14] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE
```

SUBSET OF A DATA

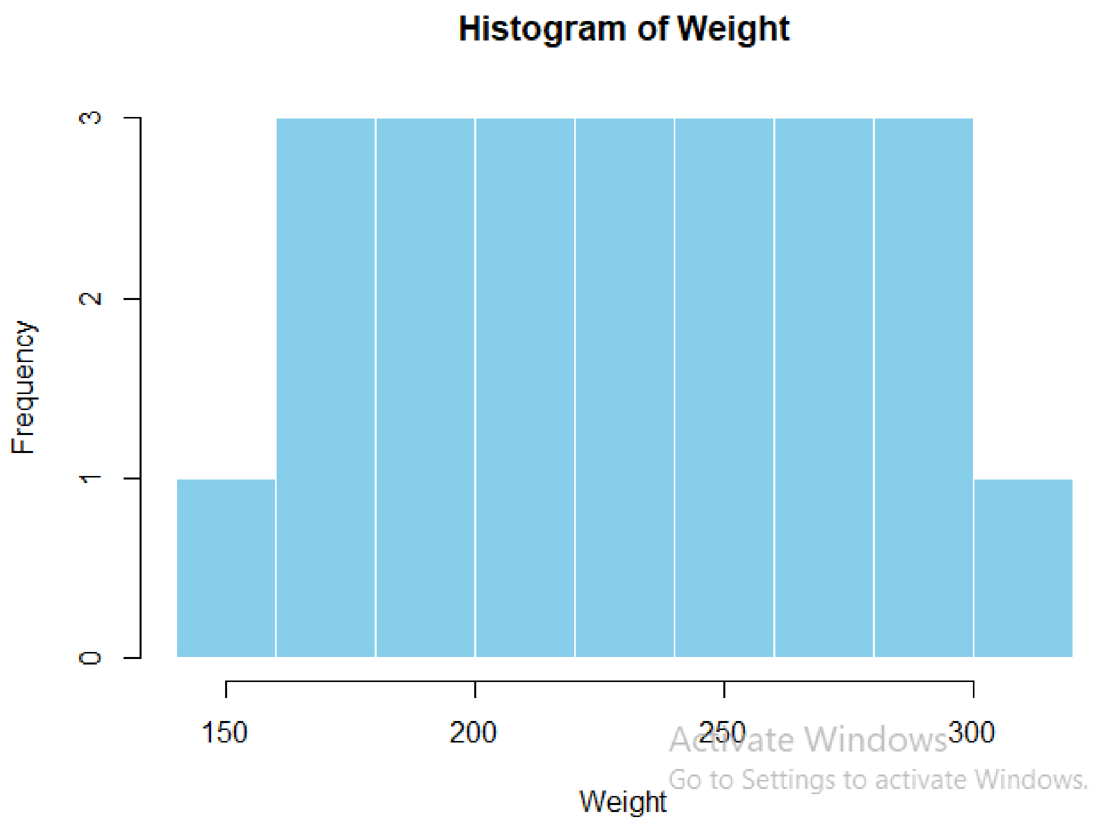
```
> df1=subset(df,Fruit_label==("Apple"))
> df1
   weight size color_score Fruit_label
6     250  8.5        0.80       Apple
7     270  8.5        0.80       Apple
8     290  9.0        0.80       Apple
12    225  8.5        0.81       Apple
13    245  8.5        0.82       Apple
14    265  8.5        0.81       Apple
15    285  8.5        0.79       Apple
> df2=subset(df,weight>200)
> df2
   weight size color_score Fruit_label
4     210  8.5        0.75      orange
5     230  8.5        0.75      orange
6     250  8.5        0.80       Apple
7     270  8.5        0.80       Apple
8     290  9.0        0.80       Apple
11    205  8.0        0.75      orange
12    225  8.5        0.81       Apple
13    245  8.5        0.82       Apple
14    265  8.5        0.81       Apple
15    285  8.5        0.79       Apple
16    305  9.0        0.92 Pomegranate
19    220  8.0        0.91 Pomegranate
20    240  7.5        0.50       lemon
21    260  7.5        0.50       lemon
22    280  8.0        0.76      orange
23    300  9.0        0.91 Pomegranate
```

SUBSET WITH RANGE

```
> df3=subset(df,weight %in% c(200:300))
> df3
   weight size color_score Fruit_label
4     210  8.5        0.75      orange
5     230  8.5        0.75      orange
6     250  8.5        0.80       Apple
7     270  8.5        0.80       Apple
8     290  9.0        0.80       Apple
11    205  8.0        0.75      orange
12    225  8.5        0.81       Apple
13    245  8.5        0.82       Apple
14    265  8.5        0.81       Apple
15    285  8.5        0.79       Apple
18    200  8.0        0.92 Pomegranate
19    220  8.0        0.91 Pomegranate
20    240  7.5        0.50       lemon
21    260  7.5        0.50       lemon
22    280  8.0        0.76      orange
23    300  9.0        0.91 Pomegranate
```

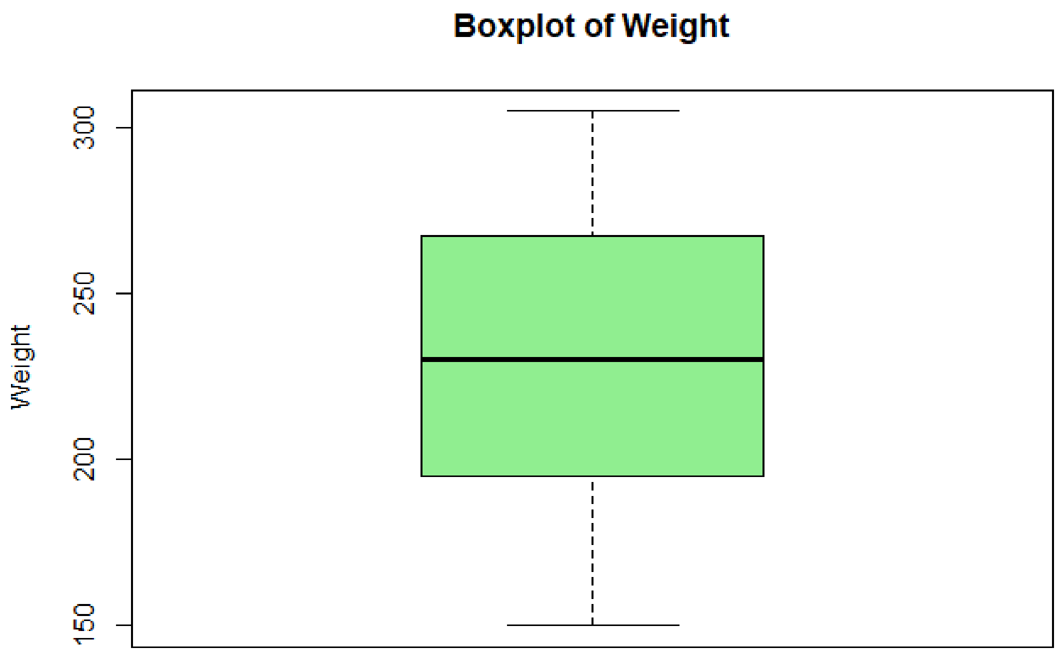
1.HISTOGRAM

```
> hist(df$weight,
+      main = "Histogram of Weight",
+      xlab = "Weight",
+      col = "skyblue",
+      border = "white")
```



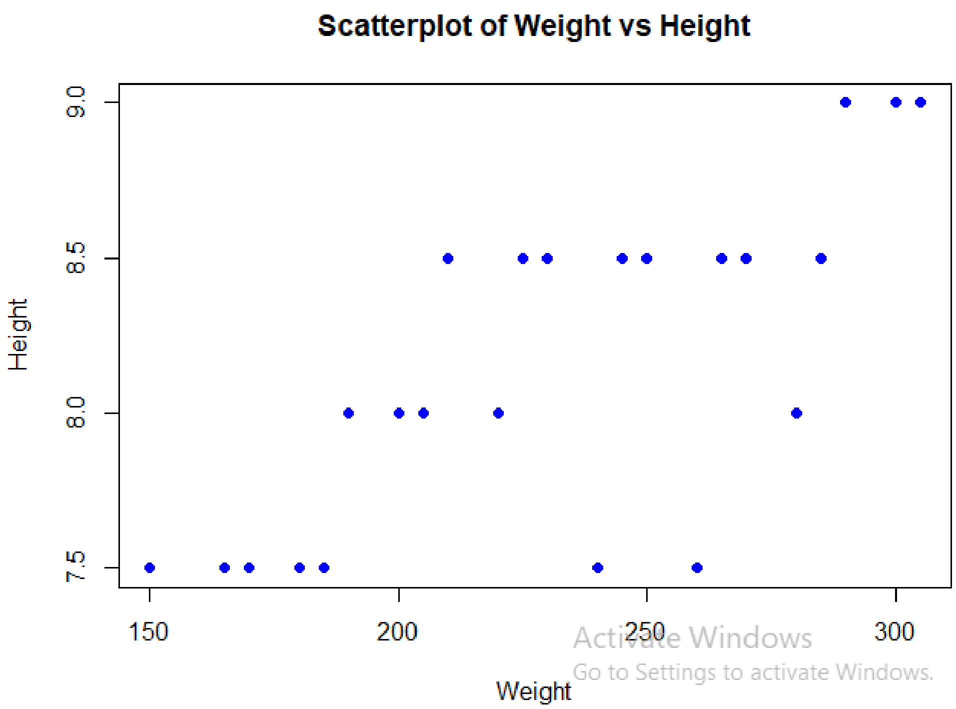
2. BOXPLOT

```
> boxplot(df$weight,
+         main = "Boxplot of Weight",
+         ylab = "Weight",
+         col = "lightgreen")
```



3. SCATTER PLOT

```
> plot(df$weight, df$size,
+      main = "Scatterplot of Weight vs Height",
+      xlab = "Weight",
+      ylab = "Height",
+      pch = 19,
+      col = "blue")
```



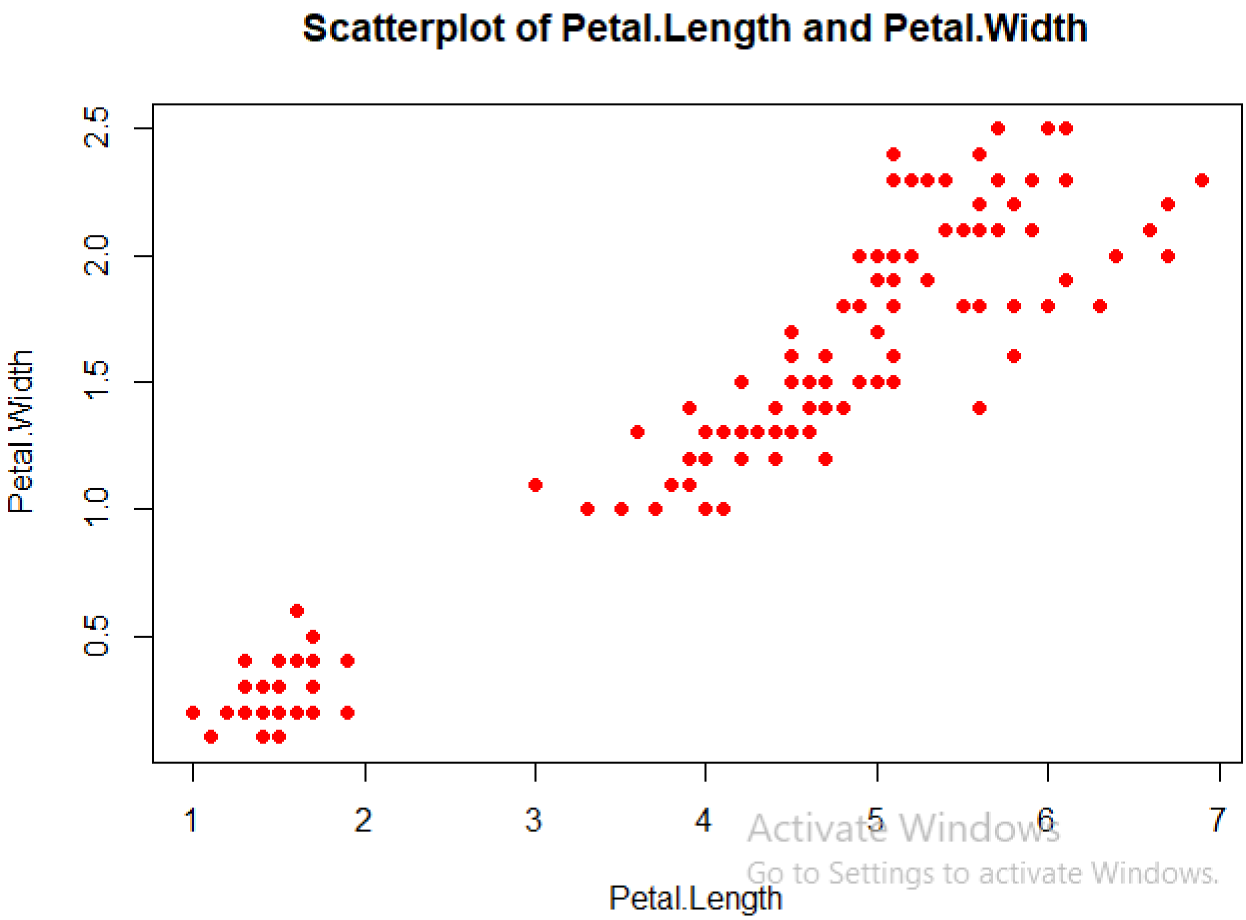
```
> #FOR IRIS DATASET
> df<-iris
> head(df)
  Sepal.Length Sepal.Width Petal.Length Petal.Width Species
1          5.1         3.5          1.4         0.2  setosa
2          4.9         3.0          1.4         0.2  setosa
3          4.7         3.2          1.3         0.2  setosa
4          4.6         3.1          1.5         0.2  setosa
5          5.0         3.6          1.4         0.2  setosa
6          5.4         3.9          1.7         0.4  setosa
> #DESCRIPTION OF DATA
> str(df)
'data.frame':	150 obs. of  5 variables:
 $ Sepal.Length: num  5.1 4.9 4.7 4.6 5 5.4 4.6 5 4.4 4.9 ...
 $ Sepal.Width : num  3.5 3 3.2 3.1 3.6 3.9 3.4 3.4 2.9 3.1 ...
 $ Petal.Length: num  1.4 1.4 1.3 1.5 1.4 1.7 1.4 1.5 1.4 1.5 ...
 $ Petal.Width : num  0.2 0.2 0.2 0.2 0.2 0.4 0.3 0.2 0.2 0.1 ...
 $ Species     : Factor w/ 3 levels "setosa","versicolor",..: 1 1 1 1 1 1 1 1 1 1 ...
> #CLASS DATATYPE
> class(df$Sepal.Length)
[1] "numeric"
> #LENGTH OF A CLASS
> length(df$Sepal.Length)
[1] 150
> #DIMENSIONS OF DATA
> dim(df)
[1] 150   5
> ncol(df)
[1] 5
> # CLASS NAMES / COLUMN NAMES
> names(df)
[1] "Sepal.Length" "Sepal.Width"  "Petal.Length" "Petal.Width"  "Species"
> #CENTRAL TENDENCY
> mean(df$Petal.Length)
[1] 3.758
> median(df$Petal.Length)
[1] 4.35
> #MEASURES OF DISPERSSION
> sd(df$Petal.Width)
[1] 0.7622377
> var(df$Petal.Width)
[1] 0.5810063
> cov(df$Petal.Length, df$Petal.Width)
[1] 1.295609
> range(df$Petal.Length)
[1] 1.0 6.9
> #SUMMARY OF A CLASS
> summary(df$Petal.Length)
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
  1.000   1.600   4.350   3.758   5.100   6.900
> summary(df)
  Sepal.Length    Sepal.Width     Petal.Length    Petal.Width          Species
 Min.   :4.300   Min.   :2.000   Min.   :1.000   Min.   :0.100   setosa    :50
 1st Qu.:5.100   1st Qu.:2.800   1st Qu.:1.600   1st Qu.:0.300   versicolor:50
 Median :5.800   Median :3.000   Median :4.350   Median :1.300   virginica :50
 Mean   :5.843   Mean   :3.057   Mean   :3.758   Mean   :1.199
 3rd Qu.:6.400   3rd Qu.:3.300   3rd Qu.:5.100   3rd Qu.:1.800
 Max.   :7.900   Max.   :4.400   Max.   :6.900   Max.   :2.500
> #TABLE
> table(df$Petal.Length>2)

FALSE  TRUE
   50   100
> table(df$Species==("versicolor"))
```

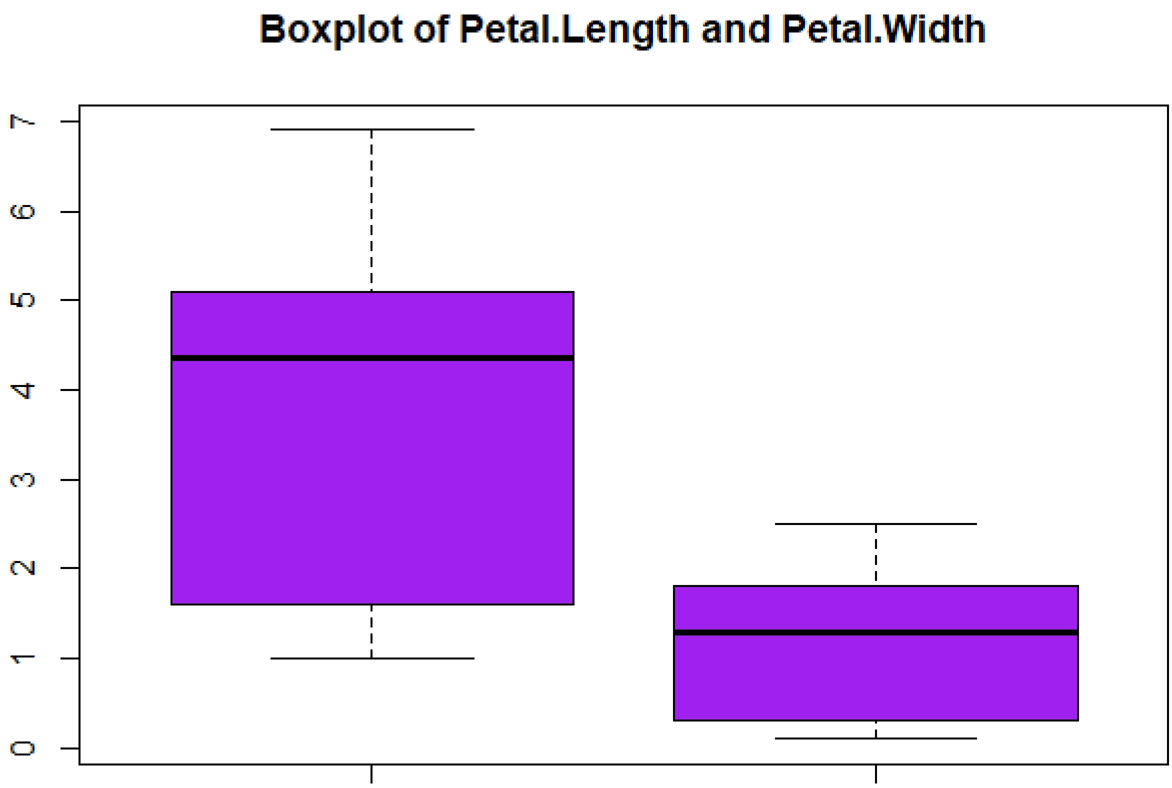
```
FALSE  TRUE 
  100    50 
> #IS NA TO FIND NULL VALUES
> is.na(df$weight)
logical(0)
> #SUBSET OF A DATA
> df1=subset(df,Species==("versicolor"))
> df1
    Sepal.Length Sepal.Width Petal.Length Petal.Width    Species
51           7.0         3.2          4.7         1.4 versicolor
52           6.4         3.2          4.5         1.5 versicolor
53           6.9         3.1          4.9         1.5 versicolor
54           5.5         2.3          4.0         1.3 versicolor
55           6.5         2.8          4.6         1.5 versicolor
56           5.7         2.8          4.5         1.3 versicolor
57           6.3         3.3          4.7         1.6 versicolor
58           4.9         2.4          3.3         1.0 versicolor
59           6.6         2.9          4.6         1.3 versicolor
60           5.2         2.7          3.9         1.4 versicolor
61           5.0         2.0          3.5         1.0 versicolor
62           5.9         3.0          4.2         1.5 versicolor
63           6.0         2.2          4.0         1.0 versicolor
64           6.1         2.9          4.7         1.4 versicolor
65           5.6         2.9          3.6         1.3 versicolor
66           6.7         3.1          4.4         1.4 versicolor
67           5.6         3.0          4.5         1.5 versicolor
68           5.8         2.7          4.1         1.0 versicolor
69           6.2         2.2          4.5         1.5 versicolor
70           5.6         2.5          3.9         1.1 versicolor
71           5.9         3.2          4.8         1.8 versicolor
72           6.1         2.8          4.0         1.3 versicolor
73           6.3         2.5          4.9         1.5 versicolor
74           6.1         2.8          4.7         1.2 versicolor
75           6.4         2.9          4.3         1.3 versicolor
76           6.6         3.0          4.4         1.4 versicolor
77           6.8         2.8          4.8         1.4 versicolor
78           6.7         3.0          5.0         1.7 versicolor
79           6.0         2.9          4.5         1.5 versicolor
80           5.7         2.6          3.5         1.0 versicolor
81           5.5         2.4          3.8         1.1 versicolor
82           5.5         2.4          3.7         1.0 versicolor
83           5.8         2.7          3.9         1.2 versicolor
84           6.0         2.7          5.1         1.6 versicolor
85           5.4         3.0          4.5         1.5 versicolor
86           6.0         3.4          4.5         1.6 versicolor
87           6.7         3.1          4.7         1.5 versicolor
88           6.3         2.3          4.4         1.3 versicolor
89           5.6         3.0          4.1         1.3 versicolor
90           5.5         2.5          4.0         1.3 versicolor
91           5.5         2.6          4.4         1.2 versicolor
92           6.1         3.0          4.6         1.4 versicolor
93           5.8         2.6          4.0         1.2 versicolor
94           5.0         2.3          3.3         1.0 versicolor
95           5.6         2.7          4.2         1.3 versicolor
96           5.7         3.0          4.2         1.2 versicolor
97           5.7         2.9          4.2         1.3 versicolor
98           6.2         2.9          4.3         1.3 versicolor
99           5.1         2.5          3.0         1.1 versicolor
100          5.7         2.8          4.1         1.3 versicolor
> df2=subset(df,Petal.Length>2)
> df2
    Sepal.Length Sepal.Width Petal.Length Petal.Width    Species
51           7.0         3.2          4.7         1.4 versicolor
```

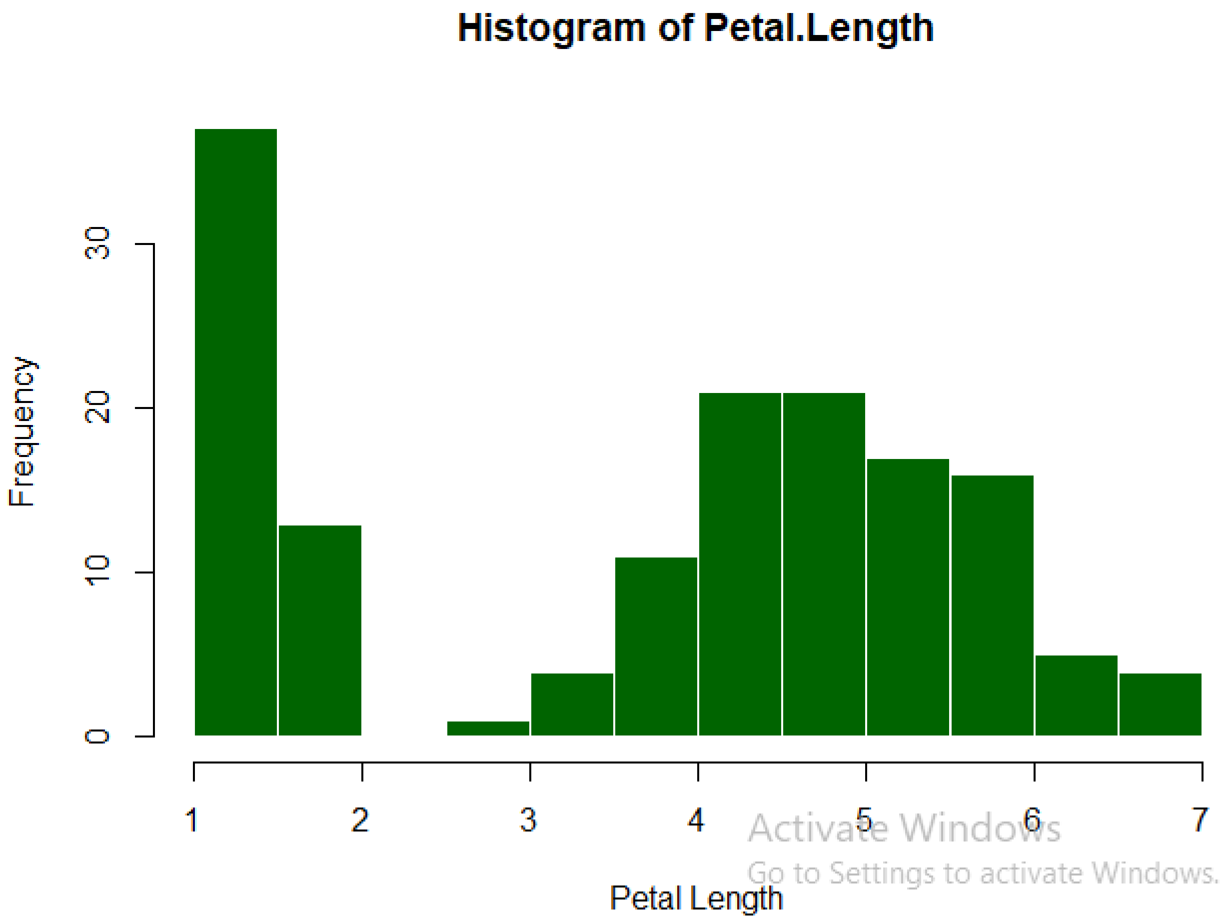
52	6.4	3.2	4.5	1.5	versicolor
53	6.9	3.1	4.9	1.5	versicolor
54	5.5	2.3	4.0	1.3	versicolor
55	6.5	2.8	4.6	1.5	versicolor
56	5.7	2.8	4.5	1.3	versicolor
57	6.3	3.3	4.7	1.6	versicolor
58	4.9	2.4	3.3	1.0	versicolor
59	6.6	2.9	4.6	1.3	versicolor
60	5.2	2.7	3.9	1.4	versicolor
61	5.0	2.0	3.5	1.0	versicolor
62	5.9	3.0	4.2	1.5	versicolor
63	6.0	2.2	4.0	1.0	versicolor
64	6.1	2.9	4.7	1.4	versicolor
65	5.6	2.9	3.6	1.3	versicolor
66	6.7	3.1	4.4	1.4	versicolor
67	5.6	3.0	4.5	1.5	versicolor
68	5.8	2.7	4.1	1.0	versicolor
69	6.2	2.2	4.5	1.5	versicolor
70	5.6	2.5	3.9	1.1	versicolor
71	5.9	3.2	4.8	1.8	versicolor
72	6.1	2.8	4.0	1.3	versicolor
73	6.3	2.5	4.9	1.5	versicolor
74	6.1	2.8	4.7	1.2	versicolor
75	6.4	2.9	4.3	1.3	versicolor
76	6.6	3.0	4.4	1.4	versicolor
77	6.8	2.8	4.8	1.4	versicolor
78	6.7	3.0	5.0	1.7	versicolor
79	6.0	2.9	4.5	1.5	versicolor
80	5.7	2.6	3.5	1.0	versicolor
81	5.5	2.4	3.8	1.1	versicolor
82	5.5	2.4	3.7	1.0	versicolor
83	5.8	2.7	3.9	1.2	versicolor
84	6.0	2.7	5.1	1.6	versicolor
85	5.4	3.0	4.5	1.5	versicolor
86	6.0	3.4	4.5	1.6	versicolor
87	6.7	3.1	4.7	1.5	versicolor
88	6.3	2.3	4.4	1.3	versicolor
89	5.6	3.0	4.1	1.3	versicolor
90	5.5	2.5	4.0	1.3	versicolor
91	5.5	2.6	4.4	1.2	versicolor
92	6.1	3.0	4.6	1.4	versicolor
93	5.8	2.6	4.0	1.2	versicolor
94	5.0	2.3	3.3	1.0	versicolor
95	5.6	2.7	4.2	1.3	versicolor
96	5.7	3.0	4.2	1.2	versicolor
97	5.7	2.9	4.2	1.3	versicolor
98	6.2	2.9	4.3	1.3	versicolor
99	5.1	2.5	3.0	1.1	versicolor
100	5.7	2.8	4.1	1.3	versicolor
101	6.3	3.3	6.0	2.5	virginica
102	5.8	2.7	5.1	1.9	virginica
103	7.1	3.0	5.9	2.1	virginica
104	6.3	2.9	5.6	1.8	virginica
105	6.5	3.0	5.8	2.2	virginica
106	7.6	3.0	6.6	2.1	virginica
107	4.9	2.5	4.5	1.7	virginica
108	7.3	2.9	6.3	1.8	virginica
109	6.7	2.5	5.8	1.8	virginica
110	7.2	3.6	6.1	2.5	virginica
111	6.5	3.2	5.1	2.0	virginica
112	6.4	2.7	5.3	1.9	virginica
113	6.8	3.0	5.5	2.1	virginica
114	5.7	2.5	5.0	2.0	virginica
115	5.8	2.8	5.1	2.4	virginica

```
116          6.4         3.2          5.3         2.3  virginica
117          6.5         3.0          5.5         1.8  virginica
118          7.7         3.8          6.7         2.2  virginica
119          7.7         2.6          6.9         2.3  virginica
120          6.0         2.2          5.0         1.5  virginica
121          6.9         3.2          5.7         2.3  virginica
122          5.6         2.8          4.9         2.0  virginica
123          7.7         2.8          6.7         2.0  virginica
124          6.3         2.7          4.9         1.8  virginica
125          6.7         3.3          5.7         2.1  virginica
126          7.2         3.2          6.0         1.8  virginica
127          6.2         2.8          4.8         1.8  virginica
128          6.1         3.0          4.9         1.8  virginica
129          6.4         2.8          5.6         2.1  virginica
130          7.2         3.0          5.8         1.6  virginica
131          7.4         2.8          6.1         1.9  virginica
132          7.9         3.8          6.4         2.0  virginica
133          6.4         2.8          5.6         2.2  virginica
134          6.3         2.8          5.1         1.5  virginica
135          6.1         2.6          5.6         1.4  virginica
136          7.7         3.0          6.1         2.3  virginica
137          6.3         3.4          5.6         2.4  virginica
138          6.4         3.1          5.5         1.8  virginica
139          6.0         3.0          4.8         1.8  virginica
140          6.9         3.1          5.4         2.1  virginica
141          6.7         3.1          5.6         2.4  virginica
142          6.9         3.1          5.1         2.3  virginica
143          5.8         2.7          5.1         1.9  virginica
144          6.8         3.2          5.9         2.3  virginica
145          6.7         3.3          5.7         2.5  virginica
146          6.7         3.0          5.2         2.3  virginica
147          6.3         2.5          5.0         1.9  virginica
148          6.5         3.0          5.2         2.0  virginica
149          6.2         3.4          5.4         2.3  virginica
150          5.9         3.0          5.1         1.8  virginica
> #SUBSET WITH RANGE
> df3=subset(df,Petal.Length %in% c(1:5))
> df3
    Sepal.Length Sepal.Width Petal.Length Petal.Width    Species
23           4.6         3.6            1         0.2     setosa
54           5.5         2.3            4         1.3 versicolor
63           6.0         2.2            4         1.0 versicolor
72           6.1         2.8            4         1.3 versicolor
78           6.7         3.0            5         1.7 versicolor
90           5.5         2.5            4         1.3 versicolor
93           5.8         2.6            4         1.2 versicolor
99           5.1         2.5            3         1.1 versicolor
114          5.7         2.5            5         2.0  virginica
120          6.0         2.2            5         1.5  virginica
147          6.3         2.5            5         1.9  virginica
> # 1.HISTOGRAM
> hist(df$Petal.Length,
+      main = "Histogram of Petal.Length",
+      xlab = "Petal Length",
+      col = "darkgreen",
+      border = "white")
> # 2. BOXPLOT
> boxplot(df$Petal.Length,df$Petal.Width,
+         main = "Boxplot of Petal.Length and Petal.Width",
+         col = "purple")
> # 3. SCATTER PLOT
> plot(df$Petal.Length, df$Petal.Width,
+      main = "Scatterplot of Petal.Length and Petal.Width",
+      xlab = "Petal.Length",
```

```
+       ylab = "Petal.Width",
+       pch = 19,
+       col = "red")
```


Scatterplot of Petal.Length and Petal.Width
Petal.Width
0.5
1.0
1.5
2.0
2.5
1
2
3
4
5
6
7
Petal.Length


Boxplot of Petal.Length and Petal.Width
0
1
2
3
4
5
6
7


Histogram of Petal.Length
Frequency
0
10
20
30
1
2
3
4
5
6
7
Petal Length
Activate Windows
Go to Settings to activate Windows.